

If It Is Funny, It Is Mean:

Understanding Social Perceptions of Yelp Online Reviews

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ABSTRACT

Online recommendation communities, like Yelp, are valuable information sources for people. Yet, we assert, review communities have their own dynamics behind the social interactions therein. In this work, we study the Yelp review votes of *useful*, *funny*, and/or *cool* to understand these social perceptions of the review. We examine the relationship between these social signals and the emotional valence of the review itself (text and rating). We aim to understand the community's perception of each of these signaling contributions. We construct a conditional inference tree of social signals from 230K Yelp reviews to study how social signals shape the deviance in review rating from the mean rating, an indicator of the overall business rating on Yelp. We find two effects of social signals. First, reviews voted as useful and funny are associated with lower user ratings and relatively negative tone in the review text. Second, reviews voted as cool tend to have a relatively positive tone and higher ratings. Our findings open a research direction for further understanding of perceptions of social signals and have implications for design of recommendation systems.

Categories and Subject Descriptors

H.4 [Information Systems Applications]: Miscellaneous;
D.2.8 [Software Engineering]: Metrics—*complexity measures, performance measures*

General Terms

Theory

Keywords

ACM proceedings, L^AT_EX, text tagging

1. INTRODUCTION

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Online recommendation sites are important resources that enable sharing opinions among people. These communities enable members (users) to share their experiences with products, services and activities in the form of reviews and ratings, that may otherwise be difficult to find before receipt of a service or product. Consumers perceive online recommendation communities as unbiased sources of information as compared to business listing websites [10]. The presence of reviews for businesses has been shown to improve customer perception of the usefulness and social presence of the website [19].

Reviews have the potential to attract user visits, increase the time spent on the site, and create a sense of community among frequent users. However, as the availability of online reviews becomes widespread, the effectiveness of recommendation communities shifts from the mere presence of customer reviews to how community members evaluate and use the reviews.

To promote engagement between the minority of content creators and the majority of content consumers, many online communities provide users with voting and feedback systems. These systems serve an important purpose—they enable users to generate *social signals* in the community e.g. Facebook's *likes*, Twitter's *favorites* and *retweets* and Amazon's *helpful review votes*. In Yelp, each review can be voted as *useful*, *cool* and/or *funny* (in any combination) which provides three social signals that describe the different ways in which users perceive online reviews written by others.

Yelp indexed over one million recommendations from 100 million unique visitors in the month of January 2013¹. Yelp members, or “Yelpers” add reviews as text with an integer rating score on a scale of one to five. The Yelpers can then vote on the review as one or more of useful, cool and funny. While the common perception is that these votes are proxies for quality of Yelp reviews, our understanding of each of these social signals is limited. In this work, we ask the question: What do these social signals for a review indicate about that review? Specifically, we ask two questions:

1. Are the social signals of a review indicative of how much the rating differs from the mean rating for that business?
2. Are the social signals of a review indicative of the polarity of the text?

We answer these questions by modeling the review's attributes (numeric rating and tone of the text) as functions of

¹<http://officialblog.yelp.com/2013/02>

the social signals which that review receives. We construct a conditional inference tree of social signals, and use review attributes as the dependent variables. We study a large-scale Yelp dataset containing 230K reviews and their social signals.

In this paper, our hypothesis is that these voting signals have specific implications towards the reviews themselves which may not match the signal’s label. To unravel such implications, we take a closer look at the relationship between these signals and the review content that is evaluated. We demonstrate two surprising observations, given that social signals for a review in Yelp are based on the review text, and have no relationship with the business that is under review. First, we find that social signals are associated with reviews whose ratings deviate from the mean rating. Second, we find that *useful* and *funny* signals are associated with reviews having a higher negative tone or lower positive tone, while *cool* signals are associated with reviews having a higher positive tone.

In the remainder of paper, we summarize related work, describe the data we obtained, and explain our statistical findings. We conclude our paper by discussing the results and describing some of the implications we can draw from this work. Finally we point out some of the limitations of this work and summarize possible future directions.

2. RELATED WORK

This paper on one hand, investigates the dynamics of behavior on online review site, Yelp, and on the other hand is focused on differentiating between various social signals communicated by users. In an effort to understand what has been done in this scope and what is missing, we summarize the relevant literature on perceptions of online reviews and the prior studies on understanding helpful votes.

2.1 Perceptions and Social Feedback

Understanding people’s perceptions of web based information systems, or social media sites is a complex phenomenon, requiring a series of systematic studies. There has been a significant amount of work on how members of online communities perceive content. Prior work has shown that attributes of review text, reviewer and social context can shape user response to reviews [2, 12, 13, 21, 27]. Lu et al. [21] use a latent topic approach to extract rated quality aspects (corresponding to concepts such as price or shipping) from comments on eBay. Harper et al. [16] studied the helpfulness of answers on the Yahoo Answers site and the influence of variables such as the type of answer and the topic domain of the question. Siersdorfer et al. [25] looked at prediction of community ratings for comments and discussions of video content on YouTube. Another study on YouTube [28] predicted the genre categorization of YouTube videos based on people’s interactions around media content without recourse to meta-data.

A body of work studied how members of online communities use review content. For example, studies have examined the role of expert reviews [6], the role of online recommendation systems [3, 5, 15], and the positive effect of social signals (and feedback mechanisms) on buyer trust [22]. Clemons et al. [7] found that positive ratings can positively influence the growth of product sales, and Chen et al. [6] found that the quality of the review as measured by helpfulness votes also positively influences sales.

A primary objective of online communities is that of user engagement. General aspects of online user engagement have been discussed in detail in prior work [20, 29]. A common way to study user engagement is through online behavior metrics that assess the extent of users’ engagement with content on the site. For instance, Peterson and Carrabis [23] describe engagement metrics that are related to consumption time on the site, users’ return to the site or subscription to the site’s feed. A study of web content by Amento et al. [1] investigated the utility of various metrics in estimating the quality of web documents. Further, Wu et al. [27] study the temporal development of product ratings and their helpfulness, and dependencies on factors such number of reviews and the effort required (writing review versus just assigning a rating).

2.2 Helpful Votes

To encourage both creators of content and the readers to engage with the site, many online communities provide users with a feedback system. Facebook likes, twitter’s favorites or retweets, Amazon’s helpful votes on reviews and Yelp’s useful, cool and funny votes are all examples of such systems. Prior research suggests that perceived attributes of the review text, reviewer and social context may all shape consumer response to reviews [2, 12, 13, 21, 27].

There is a body of work on analyzing product reviews and postings in forums. Lu et al. [21] use a latent topic approach to extract rated quality aspects (corresponding to concepts such as *price* or *shipping*) from comments in ebay. In [27] the temporal development of product ratings and their helpfulness and dependencies on factors such number of reviews or effort required (writing review vs. just assigning a rating) are studied. The helpfulness of answers on the Yahoo! Answers site and the influence of variables such as required type of answer, topic domain of the question is manually analyzed in [16]. Siersdorfer et al. [25] focused on community ratings for comments and discussions rather than product ratings on Youtube. Another study on Amazon reviews [8] studied the helpfulness scores on Amazon site and found that the helpfulness scores are not only dependent on the content of the review but also on other reviews posted for the product.

While there has been studies on understanding and predicting helpful votes, we don’t know much about other useful feedback mechanisms than can encourage engagement. Yelp provides users with two new voting concepts, funny and cool. Not only it is important to know what users in general perceive as *funny* or *cool*, but also better understandings of how social feedback system relates to conventional text analysis such as sentiment can give us insights in building more effective recommender systems.

3. DATA

Founded in 2004, Yelp is a large online recommendation community that is also a user-maintained business and service directory to help people find local businesses. According to their website², Yelp had an average of about 102 million monthly unique visitors in 2013 and the community members have written over 39 million reviews. Members of Yelp make millions of contributions to the community a month. Many regular Yelpers, maintain profiles with pictures, their

²<http://officialblog.yelp.com/>

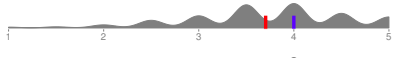
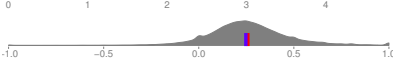
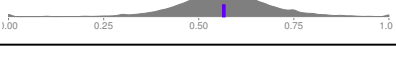
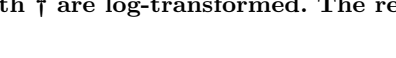
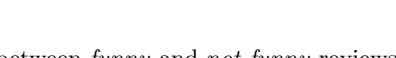
Variable	Description	Distribution
business stars	The average number of Yelp stars for the business.	
review stars	The number of stars attributed to the review. The star scale is between 1 to 5.	
review cool votes†	The number of review's cool votes.	
review funny votes†	The number of review's funny votes.	
review useful votes†	The number of review's useful votes.	
review polarity	Polarity score of review text, in range of (-1,1).	
review subjectivity	Subjectivity score of review text, in range of (0,1).	

Table 1: Distributions of variables used in this paper. Variables marked with † are log-transformed. The red and blue lines identify mean and median of the distribution respectively.

interests, favorite places, friends and location. Yelpers can friend or follow other people on site, as well as the aforementioned reviewing and voting features. We acquired our data from the Yelp Dataset Challenge³ which consists of a sample of Yelp data from the Greater Phoenix, AZ metropolitan area. It includes 11,537 *businesses*, 43,873 *reviewers* and 229,907 *reviews*. The data spans from 2005 to 2013. Table 1 summarizes the description and distributions of the numerical variables used in this paper.

As in any online recommendation community, Yelp members contribute reviews for businesses. A review consists of a stars rating and a review text. In order to understand the effect of review text, we process the text for each review to quantify the *subjectivity* (between 0 and 1) and *polarity* (between -1 and 1) of the text using the Pattern Toolkit⁴. We use the *funny*, *cool* and *useful* votes counts as measures of review's social feedback.

We construct variables for our model as follows. Our dependent variable is a binary attribute that denotes whether a review's rating was larger than the overall rating for that business (across all reviews). This measures the difference between review's rating and the average rating on Yelp for that business. We use the difference between these two ratings, as opposed to the actual review's rating to control for the average rating of the business being reviewed.

We use the *funny*, *cool* and *useful* votes as predictors of the rating deviance (defined as the difference between the review rating and the average rating for that business). Instead of using numerical values of votes for *funny*, *cool* and *useful* votes, we define a binary variable for each type of signal. The binary variable is based on whether there were more than three votes on the signal. We chose three as the threshold to control for noise and weak community feedback. We checked the results with thresholds of 0, 1, 2, 4 and 5, and found the results to be similar.

4. METHOD

In this paper, we wish to understand the relationship between each social feedback signal and the review's rating, controlling for the rating of the business. We first perform

³https://www.yelp.com/dataset_challenge

⁴<http://www.clips.ua.ac.be/pattern>

a group comparison between *funny* and *not-funny* reviews⁵, *cool* and *not-cool* reviews and *useful* and *not-useful* reviews. We use a Chi-Square (χ^2) test with the null hypothesis that both groups of *funny* reviews and *not-funny* voted reviews come from the same distribution. The test rejects the null hypothesis, indicating that they are not coming from the same distribution, with p -value smaller than 2.2×10^{-16} . We repeated this test on *cool* and *useful* votes and both tests rejected the null hypothesis with p -value smaller than 2.2×10^{-16} . Therefore, we proceed with analyzing the interaction between these social signals and understanding how they might impact the deviance in the rating.

We use conditional inference trees [17] to study the role of social feedback on deviation in a review's rating. This method is particularly useful in the case of this study, because the trees not only tell us the extent to which each variable is important, but also help us to better interpret the results. The common splitting criterion used in conditional inference trees is the Gini Index. The Gini Index checks for the "purity" of resulting children nodes in the tree. For a given node t with estimated class probabilities $p(j|t)$, $j = 1, \dots, J$ [4], the node impurity $i(t)$ is given by:

$$i(t) = \phi(p(1|t), \dots, p(J|t))$$

The most favorable split is the one that reduces the node or equivalently the tree impurity. The most favorable split is identified by search. Adopting the Gini diversity index, $i(t)$ takes the form:

$$i(t) = \sum_{j \neq i} p(j|t)p(i|t)$$

The Gini index, defined as a function $\phi(p_1, \dots, p_J)$ of the $p_1, \dots, p_J$, is a quadratic polynomial with nonnegative coefficients. In conditional inference trees, the node split is selected based on how good the association is. The resulting node should have a higher association with the observed value of the dependent variable. The conditional inference tree uses a χ^2 test to test the association. Therefore, it not only removes the bias due to categories but also chooses those variables that are informative.

The key to this algorithm is based on the separation of variable selection and the splitting procedure. The recursive

⁵We consider a review to be funny if it has at least 4 funny votes (see Section 3).

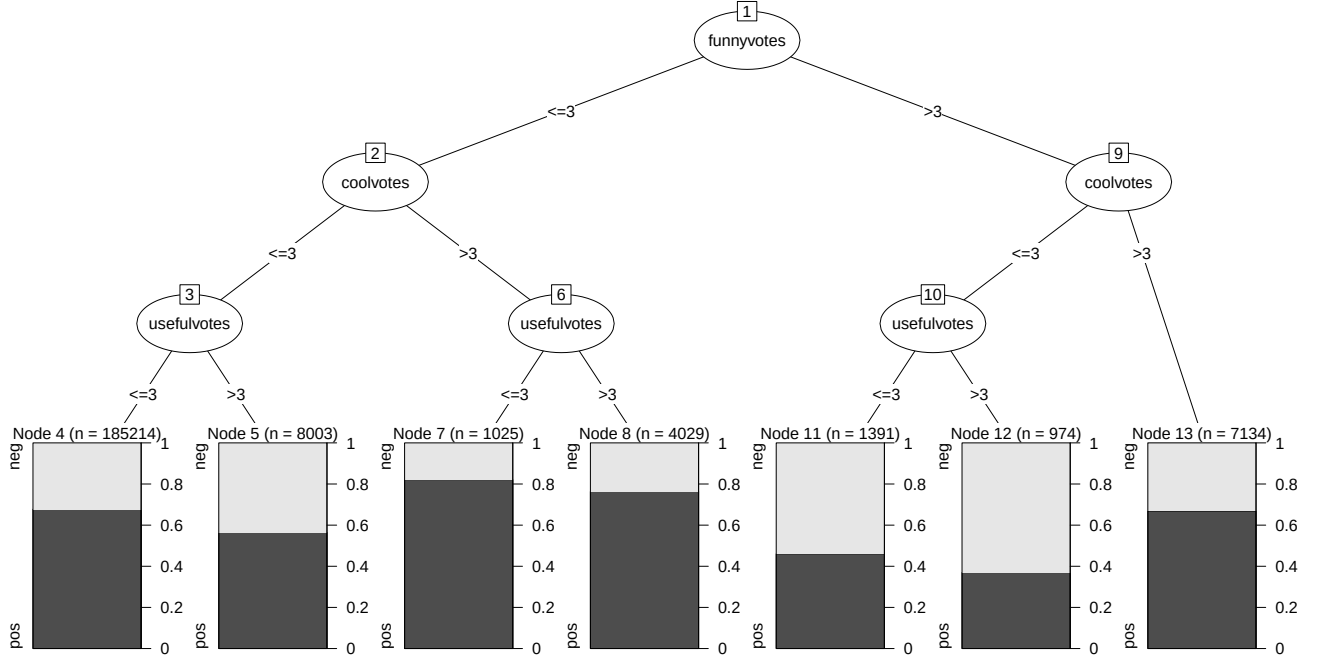


Figure 1: Conditional inference tree with difference in review’s rating and business’s rating as dependent variable. For each inner node, the Bonferroni-adjusted p -values for all splits are less than 0.001, the number of data points in each group is displayed for each terminal node. The black color in the bar plots quantifies the likelihood of the review’s rating being greater than the business review, and the gray area shows the likelihood of the review’s rating being lower than the business’s collective average rating.

binary partitioning is as follows. The response Y comes from sample space Y , which may be multivariate. The m dimensional covariate vector $X = (X_1, \dots, X_m)$ is taken from a sample space $X = X_1 \times \dots \times X_m$. Both the response variables and the dependent variable may be measured at any arbitrary scale. The conditional distribution of the response variable given the covariates depends on the function of covariates:

$$\begin{aligned} D(Y|X) &= D(Y|X_1, \dots, X_m) \\ &= D(Y|f(X_1, \dots, X_m)) \end{aligned}$$

For a given learning sample of n iid observations a generic algorithm can be formulated using nonnegative integer of case weights $w = (w_1, w_2, \dots, w_n)$. Each node of a tree is represented by a vector of case weights having non-zero elements when the corresponding observations are significant for that node, and zero otherwise. The generic algorithm is:

1. For case weights w , we test the global null hypothesis of independence between any of the covariates and the response. The step terminates if the hypothesis cannot be rejected at a pre-specified nominal level α . Otherwise, we select the j ’th covariate X_j with the strongest association to the response variable.
2. Set $A \subset X_j$ is chosen to split X_j , into two disjoint sets. The case weights w_{left} and w_{right} determine the two subgroups with

$$w_{left,i} = w_i I(X_{ji} \in A)$$

and

$$w_{right,i} = w_i I(X_{ji} \notin A)$$

for all $i = 1, \dots, n$, where $I(\cdot)$ denotes the indicator function, which indicates the membership of an element in a subset.

3. We repeat the steps 1 and 2 with modified case weights w_{left} and w_{right} , respectively.

The separation of variable selection and splitting procedures is essential for the development of trees with no tendency toward covariates with many possible splits [17].

5. RESULTS

Figure 1 visualizes the conditional inference tree for our dependent variable (the difference in review’s rating and the business’s rating). As we can see in Figure 1, the terminal nodes show different distributions of the dependent variable (the deviance from mean rating) in each case. From Figure 1 we can see that on the right side of the tree where the number of *funny* votes is greater than three, there is higher density of negative deviance from the mean. This means the reviews written for the business with more *funny* votes are usually rated lower than the mean business rating. In other words, the *funny* votes are usually given to the more critical reviews that have lower ratings. The negative tendency of the ratings compared to the collective business rating shows that the concept of *funny* is perceived negatively by the Yelp users. Table 2 summarizes the mean, standard deviation and

standard error of sentiment polarity for reviews considered *funny* or *not-funny*. We can see that the funny reviews are -0.39 units more negative than the not-funny ones.

We find that when the *funny* votes are high but the *cool* votes are low (Figure 1), we see the most negative deviance in the ratings from the mean business’s rating. When the useful votes are low as well, the likelihood of having a negative deviance rating is equal to 0.5 but when the useful votes are high, the negative deviance likelihood is as high as 0.6, meaning that 60% of the reviews with higher *funny* votes, lower *cool* votes and higher *useful* votes are rated lower than the collective business’s rating. We show an example of parts of a review that was voted funny and useful but did not receive many cool votes:

“...Upon arrival, we were immediately greeted by a smell that can accurately be described as ‘Fat, grease, and charred death.’ ...

As far as decor, the restaurant does a good job of sticking to an anti-doctor theme, with signs plastered all over the restaurant warning you of the unhealthy nature of their food, mandatory patient gowns for all diners, and waitresses dressed as cute nurses.

But the food itself could be summed up with the word ‘Sickening,’ which actually works metaphorically and literally. Their shakes are incredibly sweet, but actually filled with ice cream, pure cream, and butter. Their fries are cooked in lard, and their burgers are excessively heavy and fattening. A burger and fries is really enough to fill up a fully-grown man, but the sensation of being completely filled by lard, meat, and grease is almost as disgusting as seeing one of the obese regulars scarf down a triple-bypass burger for free. ...

Disgusting? Yes. Necessary? No. Immoral? Decidedly no, since they clearly warn you about what you’re getting in to.

I’d eat here once, and NEVER again. It’s a place fueled by novelty- go there only if you’re inspired after watching Epic Meal Time on youtube.”

(2 cool votes, 17 funny votes, 11 useful votes)

While *funny* votes are given to more critical reviews (lower ratings), the *cool* votes seem to enforce a positive perception on the users. On the other hand the *useful* votes when accompanied by *funny* votes, seem to be related to lower ratings. We see that in the case of high *funny* votes and high *cool* votes the likelihood of a review rating being smaller than the collective business review is 0.3, see Figure 1.

When the *funny* votes are low and *cool* votes are high, we see the highest likelihood of positive change in ratings. In the case that *useful* votes are low, the likelihood of positive change is the highest among all, i.e. 0.8. When the *useful* votes are high, the likelihood is slightly lower but still close to 0.8. This shows the positive perception of *cool* votes and negative perception of *funny* votes. An example review, where the cool votes were high and useful and funny votes were low is provided below:

“Ode to Trader Joe’s

Seasonal produce of mellow fruitfulness!

Condition	Mean Polarity	Std.Dev	Std.Err
Funny > 3	-0.17	0.16	0.002
Funny ≤ 3	0.26	0.22	0.000
Cool > 3	0.46	0.22	0.001
Cool ≤ 3	0.21	0.15	0.001
Useful > 3	0.09	0.16	0.001
Useful ≤ 3	0.26	0.22	0.001

Table 2: Mean, Standard Deviation and Standard Error values of polarity for reviews voted with each of the social feedback signals. We can see the mean polarity for funny reviews is around 0.4 less than not-funny ones. The cool reviews are more positive in 0.25 units in their polarity and the useful votes are 0.17 less positive than the not-useful ones.

Island themed friend of the tropical sun;
Conspired to load the shopping cart
With fruit and veggies that round the table run;
Oft whoever seeks tasty may find. ...

Where are thy Joe-O’s? Ay, where are they?
Think of them, and thy hast cheeses too,
Then in a clanging call the young cashiers be-
seach
Open yet another among the checkout stands
And gathering shoppers swoon in the lot.

With a nod to Keats”
(11 cool votes, 1 funny votes, 3 useful votes)

Table 2 shows that the difference among the sentiment polarity of cool reviews and the ones without *cool* votes is also significant (0.25), indicating that both the ratings and polarity of the reviews are higher among the ones with cool votes.

The *useful* votes are also slightly negatively perceived. When the *funny* votes are low and the *cool* votes and *useful* votes are low as well, we have the most common case happening ($n = 185214$). In this case, the chance of negative change is 0.3 and the chance of positive deviance is 0.7. We can see from Figure 1 that in the case of low *funny* and *cool* votes but higher *useful* votes, the chances of negative change increases to 0.4, indicating that the perception of *useful* signal on Yelp is negative. The polarity of the sentiments on *useful* reviews is around 0.17 units less than the ones without *useful* votes, which emphasizes on the negative tone of useful reviews.

6. DISCUSSION

In this paper, we have shown how different social signals, used as feedback votes, demonstrates the differences of review ratings and their sentiments. We started with a limited understanding of what users perceive as *cool*, *funny* or *useful* and took a first step towards understanding the relative relationship of each of these social signals with respect to the changes in the review ratings. We were able to point out significant differences in review rating patterns and emotional

valence of the review text across these social signals. We summarize our findings and discuss implications below.

Funny votes imply lower ratings and higher negativity in the tone.

Our results show that the funny votes are related to higher likelihood of negative changes in the ratings. This means that the population of Yelp users perceive those reviews as funny that are lower in ratings. This can be explained in two ways. First, it might be that the general audience on the review sites such as Yelp, enjoy reading the reviews with sarcastic tone or those reviews that criticize the business with some humor. The second explanation goes back to the perception of the reader and the votes. It could be that the users find the lower rated reviews and those with negative tone more funny or humorous. Either way, the fact that the higher funny votes are related to the lower rated reviews illuminates the opinion of the general Yelp user of what is considered funny. Further, it's worth mentioning that negativity of funny reviews can help with crowd-generated content that may be identified as sarcasm. Previous research has investigated ways to detect sarcasm through natural language models [9, 26].

Useful votes imply lower ratings and lower positivity in the tone.

The same pattern that is observable with funny votes also applies to useful votes, but in a smaller scale. Our results show that the reviews that are rated as useful are on average more likely to be rated lower than those without useful votes. This finding suggests that users find the more critical reviews more useful. While, useful reviews are usually perceived as high quality reviews on review sites, here our findings emphasizes its relationship with lower ratings and higher negative sentiment.

Cool votes imply higher ratings and higher positivity in the tones.

Unlike the perceptions of funny and useful on the reviews, we find that reviews that are perceived as cool are more likely to be highly rated. This suggests that the cool perception is usually tied with the higher rated reviews of businesses. The lower rated reviews are less likely to be perceived as cool. We also find that the level of polarity in the review is higher in positivity among cool reviews, which makes the argument of positive perception of cool reviews stronger.

Community signals carry implied meaning beyond their specific labels.

Online recommendation communities rely on member contributions to index and serve recommendations to their users [11, 14, 18]. The findings of our research suggest that there is deeper meanings and interaction forms than the generic votes. The specific community of Yelp, for example, judges and communicates meaning through its own interpretation of the signal labels, which is congruent to similar finding in the multimedia community [24]. Future work can elaborate on the use of these labels and their corresponding social perceptions in other communities. For example, what do *likes* on Facebook mean in different contexts and by different people? Is it some form of social confirmation, do users support the content of the post? Do they find it cool or funny? Deeper

understandings of forms of user feedbacks can enable us with better recommendation and social network design.

This study is based on Yelp and further work is needed to ascertain whether our findings apply to other review communities. For example the negative tone in funny reviews might not be appreciated in other types of communities. Further, our data was limited to certain variables and the statistical methods we used examine only a small segment of behavior on Yelp. A possible direction for future work is to look closely into the language that is used in funny, cool and useful reviews. For example, are there certain phrases or types of writing that contribute to funny votes? Another possible research direction is to complement this work with a qualitative study. One could find what users perceive as cool or funny by interviewing Yelp users.

7. CONCLUSION

Online recommendation communities such as Yelp enable people to find right businesses and services, and enable businesses to find new customers. Hence, it is critical for such communities to maximize social interactions around reviews. Yelp, as a pioneer in online review communities, has provided users with several ways to evaluate content. In this paper, we took a first step to better understand the underlying review patterns associated with each of these social signals. We found that useful and funny signals are mostly associated with lower ratings and more negative sentiment, while cool signal is associated with higher ratings and positive sentiment.

8. REFERENCES

- [1] B. Amento, L. Terveen, and W. Hill. Does “authority” mean quality? predicting expert quality ratings of web documents. In *Proceedings of the 23rd annual international ACM SIGIR conference on Research and development in information retrieval*, pages 296–303. ACM, 2000.
- [2] N. Archak, A. Ghose, and P. G. Ipeirotis. Show me the money!: deriving the pricing power of product features by mining consumer reviews. In *Proceedings of the 13th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 56–65. ACM, 2007.
- [3] J. Y. Bakos. Reducing buyer search costs: implications for electronic marketplaces. *Management science*, 43(12):1676–1692, 1997.
- [4] L. Breiman. *Classification and regression trees*. CRC press, 1993.
- [5] P.-Y. Chen, S.-y. Wu, and J. Yoon. The impact of online recommendations and consumer feedback on sales. 2004.
- [6] Y. Chen and J. Xie. Online consumer review: Word-of-mouth as a new element of marketing communication mix. *Management Science*, 54(3):477–491, 2008.
- [7] E. K. Clemons, G. G. Gao, and L. M. Hitt. When online reviews meet hyperdifferentiation: A study of the craft beer industry. *Journal of Management Information Systems*, 23(2):149–171, 2006.
- [8] C. Danescu-Niculescu-Mizil, G. Kossinets, J. Kleinberg, and L. Lee. How opinions are received by online communities: a case study on amazon. com helpfulness votes. In *Proceedings of the 18th international conference on World wide web*, pages 141–150. ACM, 2009.
- [9] D. Davidov, O. Tsur, and A. Rappoport. Semi-supervised recognition of sarcastic sentences in twitter and amazon. In *Proceedings of the Fourteenth Conference on Computational Natural Language Learning*, pages 107–116. Association for Computational Linguistics, 2010.
- [10] C. Dellarocas. The digitization of word of mouth: Promise and challenges of online feedback mechanisms. *Management science*, 49(10):1407–1424, 2003.
- [11] K. E. Finn, A. J. Sellen, and S. B. Wilbur. *Video-mediated communication*. L. Erlbaum Associates Inc., 1997.
- [12] C. Forman, A. Ghose, and B. Wiesenfeld. Examining the relationship between reviews and sales: The role of reviewer identity disclosure in electronic markets. *Information Systems Research*, 19(3):291–313, 2008.
- [13] A. Ghose, P. G. Ipeirotis, and A. Sundararajan. Opinion mining using econometrics: A case study on reputation systems. 45(1):416, 2007.
- [14] A. Girgensohn and A. Lee. Making web sites be places for social interaction. In *Proceedings of the 2002 ACM conference on Computer supported cooperative work, CSCW ’02*, pages 136–145, New York, NY, USA, 2002. ACM.
- [15] U. Gretzel and D. R. Fesenmaier. Persuasion in recommender systems. *International Journal of Electronic Commerce*, 11(2):81–100, 2006.
- [16] F. M. Harper, D. Raban, S. Rafaeli, and J. A. Konstan. Predictors of answer quality in online q&a sites. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 865–874. ACM, 2008.
- [17] T. Hothorn, K. Hornik, and A. Zeileis. Unbiased recursive partitioning: A conditional inference framework. *Journal of Computational and Graphical Statistics*, 15(3), 2006.
- [18] J. Koh, Y.-G. Kim, B. Butler, and G.-W. Bock. Encouraging participation in virtual communities. *Commun. ACM*, Feb. 2007.
- [19] N. Kumar and I. Benbasat. Research note: the influence of recommendations and consumer reviews on evaluations of websites. *Information Systems Research*, 17(4):425–439, 2006.
- [20] J. Lehmann, M. Lalmas, E. Yom-Tov, and G. Dupret. Models of user engagement. In *User Modeling, Adaptation, and Personalization*, pages 164–175. Springer, 2012.
- [21] Y. Lu, C. Zhai, and N. Sundaresan. Rated aspect summarization of short comments. In *Proceedings of the 18th international conference on World wide web*, pages 131–140. ACM, 2009.
- [22] P. A. Pavlou and A. Dimoka. The nature and role of feedback text comments in online marketplaces: Implications for trust building, price premiums, and seller differentiation. *Information Systems Research*, 17(4):392–414, 2006.
- [23] E. T. Peterson and J. Carrabis. Measuring the immeasurable: Visitor engagement. *Web Analytics Demystified*, 2008.
- [24] D. A. Shamma, R. Shaw, P. L. Shafton, and Y. Liu. Watch what i watch: using community activity to understand content. In *MIR ’07: Proceedings of the international workshop on Workshop on multimedia information retrieval*, pages 275–284, New York, NY, USA, 2007. ACM.
- [25] S. Siersdorfer, S. Chelaru, W. Nejdl, and J. San Pedro. How useful are your comments?: analyzing and predicting youtube comments and comment ratings. In *Proceedings of the 19th international conference on World wide web*, pages 891–900. ACM, 2010.
- [26] O. Tsur, D. Davidov, and A. Rappoport. Icwsma great catchy name: Semi-supervised recognition of sarcastic sentences in online product reviews. In *ICWSM*, 2010.
- [27] F. Wu and B. A. Huberman. How public opinion forms. In *Internet and Network Economics*, pages 334–341. Springer, 2008.
- [28] J. Yew, D. A. Shamma, and E. F. Churchill. Knowing funny: genre perception and categorization in social video sharing. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 297–306. ACM, 2011.
- [29] E. Yom-Tov, M. Lalmas, R. Baeza-Yates, G. Dupret, J. Lehmann, and P. Donmez. Measuring inter-site engagement. *IEEE International Conference on Big Data*, 2013.